

Machine Learning for Credit Card Fraud Detection

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1 Introduction

Fraud detection has become an essential aspect of ensuring financial security in modern digital payment systems. With estimated global losses due to fraud exceeding billions of dollars annually, addressing this challenge has become a priority for financial institutions, businesses, and consumers alike. Beyond financial implications, successful fraud detection systems foster trust in digital payment platforms, encouraging their adoption and enhancing user confidence.

Current fraud detection systems rely heavily on rule-based approaches and machine learning models. Rule-based systems are rigid and fail to adapt to evolving fraud patterns, while machine learning models often struggle with data imbalance, leading to biased predictions. Moreover, the opacity of complex models like deep learning creates trust issues for stakeholders seeking interpretable solutions.

The objective of this project was to develop a machine learning model capable of identifying fraudulent transactions in a highly imbalanced dataset. Logistic Regression was chosen for its interpretability and effectiveness in binary classification tasks. By addressing the imbalance issue and optimizing model performance, the project aimed to create a reliable, efficient, and interpretable fraud detection system.

Fraud detection in its current form involves the use of machine learning and rule-based systems. However, these approaches face significant limitations, including:

- **Class Imbalance:** Fraudulent transactions are extremely rare, leading to biased models.
- **False Negatives:** Missing fraudulent transactions results in financial losses.
- **False Positives:** Flagging legitimate transactions inconveniences customers.
- **Lack of Interpretability:** Complex models such as deep learning often lack transparency.

A successful fraud detection system reduces financial loss, enhances customer trust, and ensures the integrity of digital payment systems. By leveraging logistic regression and incorporating advanced techniques such as class balancing and threshold optimization, this project demonstrates a novel approach to tackling fraud detection challenges.

2 Methodology and Results

2.1 Data Preprocessing and Balancing

Data preprocessing was a critical step to ensure that the model could handle the inherent challenges of the dataset. First, the dataset was examined to understand its characteristics. It was found that

fraudulent transactions constituted only 0.172% of the dataset. This severe imbalance necessitated the use of class balancing techniques.

Borderline-SMOTE was employed to generate synthetic samples near the decision boundaries. This method was chosen over traditional SMOTE as it better preserves the structure of the data and focuses on the most challenging cases for classification.

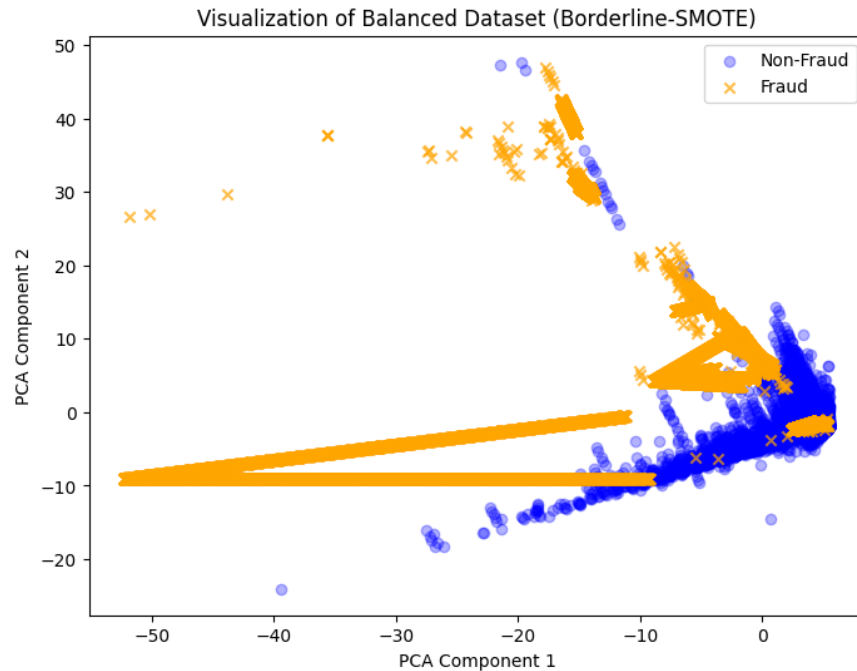


Figure 1: Visualization of Balanced Dataset Using Borderline-SMOTE. Borderline-SMOTE focuses on the decision boundary, improving the model’s ability to classify borderline cases.

The PCA visualization shows how Borderline-SMOTE improves the separability between fraudulent and legitimate transactions, particularly for borderline cases.

2.2 Model Training and Evaluation Metrics

Logistic Regression was selected due to its simplicity, efficiency, and interpretability. It is particularly well-suited for binary classification tasks such as fraud detection. To enhance the model’s performance, Stochastic Gradient Descent (SGD) was used for optimization. Regularization, specifically L2 (Ridge), was incorporated to prevent overfitting and to handle the high-dimensional feature space effectively.

Cost-sensitive learning was integrated by adjusting class weights to penalize misclassification of fraudulent transactions more heavily. This approach ensured that the model remained sensitive to rare fraud cases without compromising overall accuracy.

Evaluation metrics used to measure the model’s performance include:

- **Precision:** The proportion of true positives among all predicted positives.
- **Recall:** The proportion of true positives among all actual positives.

- **F1-Score:** The harmonic mean of precision and recall.
- **ROC AUC:** The area under the Receiver Operating Characteristic curve, summarizing the model's ability to distinguish between classes.

2.3 Threshold Tuning and Precision-Recall Curve

Threshold tuning played a vital role in balancing precision and recall. While the default decision threshold for Logistic Regression is 0.5, this project optimized the threshold to maximize the F1-Score. By experimenting with various thresholds, an optimal threshold of 62.14 was identified. This adjustment significantly reduced false negatives (missed fraud cases) while maintaining a reasonable false positive rate.

To better understand the trade-off between precision and recall, a Precision-Recall Curve was generated. This curve illustrates how the model's performance changes with varying decision thresholds.

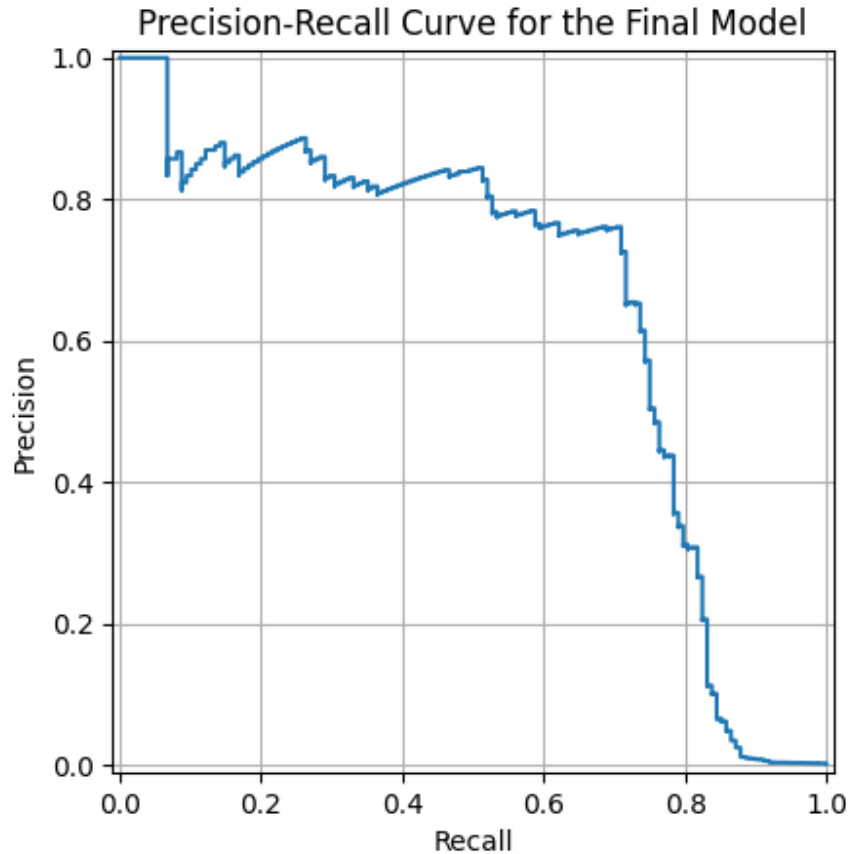


Figure 2: Precision-Recall Curve for the Final Model. The annotated threshold (62.14) achieves the best balance between precision and recall, optimizing the F1-Score.

2.4 Comparison of Class Weights

Class weights were adjusted during model training to address data imbalance. The comparison below highlights the impact of different class weight configurations on model performance.

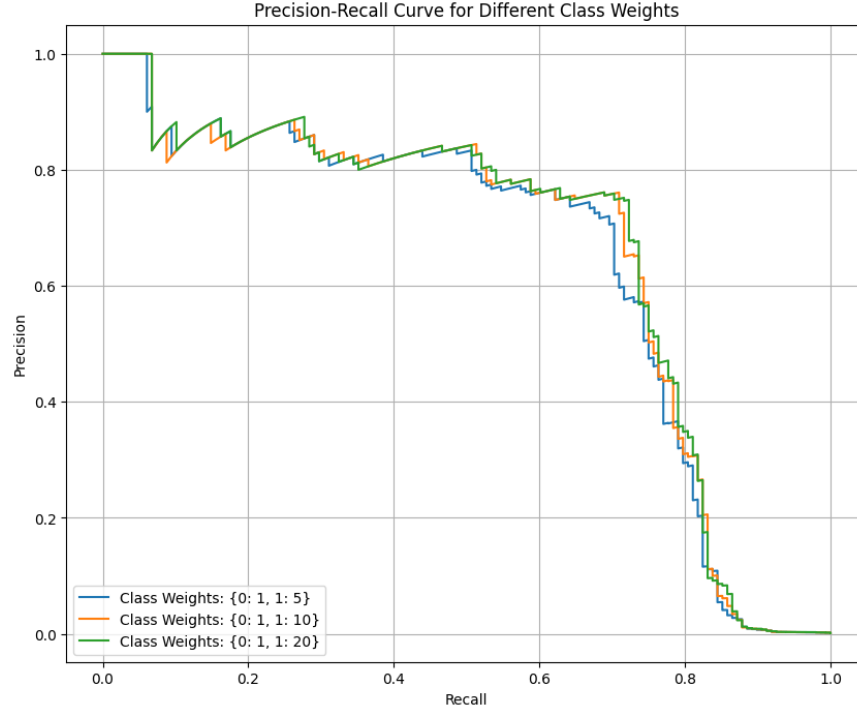


Figure 3: Performance Comparison for Different Class Weights. Class weight $\{0: 1, 1: 10\}$ provided the best balance between precision and recall.

Class Weights	Optimal Threshold	Precision	Recall	F1-Score
$\{0: 1, 1: 5\}$	43.54	0.72	0.69	0.70
$\{0: 1, 1: 10\}$	62.14	0.76	0.70	0.73
$\{0: 1, 1: 20\}$	100.07	0.75	0.72	0.73

The configuration `class_weight={0: 1, 1: 10}` provided the best balance between precision and recall, achieving the highest F1-Score without significant trade-offs.

2.5 Final Results and Confusion Matrix Analysis

After optimizing the threshold and using `class_weight={0: 1, 1: 10}`, the final model achieved the following metrics:

- **Precision:** 76%
- **Recall:** 70%
- **F1-Score:** 73%

- **ROC AUC:** 94.4%

The confusion matrix provides a detailed breakdown of the model’s predictions, highlighting true positives, true negatives, false positives, and false negatives.

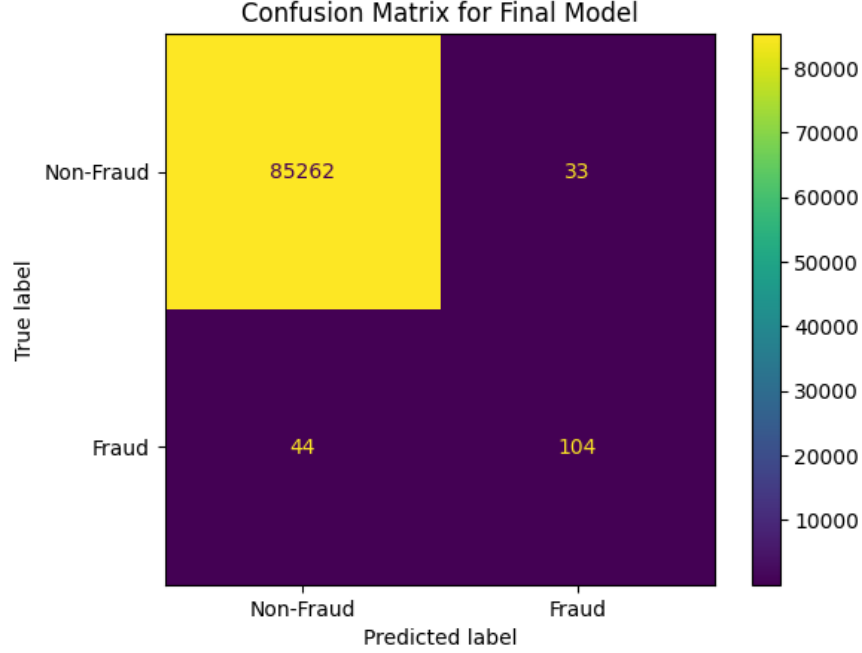


Figure 4: Confusion Matrix for Final Model. True negatives and true positives are maximized, while false negatives are minimized, demonstrating effective fraud detection.

3 Availability and Reproducibility

The source code and dataset preprocessing scripts are included in the submission. To ensure reproducibility, we have included the training, validation, and test datasets, along with a fixed random seed (`random_state=42`) to replicate the results. Model parameters, including the final optimized class weights and threshold, are detailed in the accompanying configuration file.

4 Conclusion

This project successfully developed a robust and interpretable fraud detection system using Logistic Regression. The integration of Borderline-SMOTE, cost-sensitive learning, and threshold optimization addressed the challenges posed by class imbalance and improved detection of rare fraudulent transactions. The final model achieved high performance metrics, including an F1-Score of 73% and a ROC AUC of 94.4%. These results demonstrate that Logistic Regression, when combined with thoughtful preprocessing and optimization, can effectively address the challenges of fraud detection.

Future Work

While this project focused on interpretability and performance with Logistic Regression, several promising directions could further improve fraud detection:

- **Ensemble Methods:** Algorithms such as Random Forests, Gradient Boosted Trees (e.g., XGBoost), and bagging could improve performance by combining multiple weak learners.
- **Deep Learning Models:** Neural networks, especially with recurrent architectures (e.g., LSTMs) or attention-based models (e.g., Transformers), could capture sequential patterns in transaction histories.
- **Real-Time Detection:** Developing and deploying models for streaming data could make fraud detection systems more responsive in production environments.
- **Explainability Tools:** Incorporating model interpretability techniques such as SHAP or LIME would allow stakeholders to better understand why transactions are flagged as fraudulent.
- **Feature Engineering:** Creating time-aware or user-specific features based on transaction patterns could improve the model’s ability to detect anomalous behavior.
- **Unsupervised and Semi-Supervised Learning:** Given the scarcity of labeled fraud cases, exploring anomaly detection and self-supervised methods could uncover unseen fraud without relying on labeled data.

5 Works Cited

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